Do Now · retrieval

Lesson 073 · Trace Tables and Boundary Testing

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What does CountPasses count?

Q2

What is a trace table for?

Q3

What is a boundary condition?

Q4

Stack Push 3, Push 7, then Pop. What is popped and what remains?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Observe — A trace table records variable values after each meaningful execution step: i, a[i], condition and result.

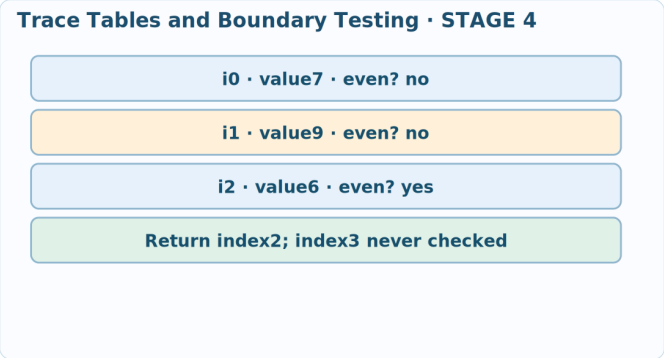
Core idea 2

Explain — A boundary test probes just below, at and above a threshold. Normal, erroneous and exceptional tests reveal different faults.

Core idea 3

Evaluate — Test early success, late success, absent target, and empty arrays. For this function, −1 distinguishes not found from index0.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

trace table — State changes in algorithm execution.

sentinel — Special value indicating a condition.

boundary test — Values at or near limit.

exceptional test — Unusual or error-causing case.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

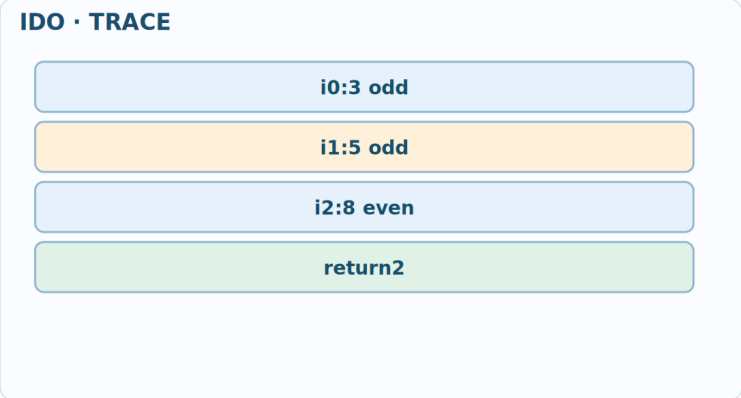
Problem

Trace first even index in [3,5,8,10].

Follow the worked method

1. Explain — The function exits at first even occurrence and does not examine10.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Explain
No candidate satisfies value%2==0, so the not-found sentinel is returned.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Trace FindFirstEven([7,9,6,4]) with index and parity. [6]

Question 2

Identify expected outputs for [], [2,3], [3,5,8], [1,7,9]. [4]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A program incorrectly returns0 when there is no even value. Explain the ambiguity and propose a safer fix, with tests. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Returning0 is a safe 'not found' value for zero-indexed arrays.

Use key vocabulary from memory

1. Explain trace table in your own words.

2. Explain sentinel in your own words.

3. Explain boundary test in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.